Helia Vafaei

Tehran **+98 (991) 290-9125** **helia.vafaei2080@gmail.com**

LINKS LinkedIn, GitHub, Personal website

EDUCATION

Sep 2025 – Present
M.Sc. in Information Technology (IT) Engineering
Major: Networked Systems
University of Tehran Tehran
Current GPA: **17.24/20**

Sep 2020 – Feb 2025
B.Sc. in Computer Engineering
Iran University of Science and Technology Tehran
GPA: **17.27/20** | Final two years: **17.56/20**

RESEARCH INTERESTS

Software Engineering
AI for Software Engineering
Graph Machine Learning and Network Science
AI for Healthcare

RESEARCH EXPERIENCE

University of Tehran | Sep 2026 – Present
M.Sc. Research — Dynamic Functional Brain Network Analysis for MCI Classification

- Investigating dynamic functional brain networks from resting-state fMRI (rs-fMRI) for the analysis and classification of Mild Cognitive Impairment (MCI).
- Curating and preprocessing ADNI rs-fMRI and T1-weighted MRI data using reproducible neuroimaging workflows based on BIDS and fMRIPrep.
- Exploring functional connectivity, graph-based representations, and machine-learning approaches for identifying patterns associated with cognitive impairment.

PUBLICATIONS

H. Vafaei, N. Haghighi, and M. Malekimajd, "Intelligent Resource Allocation in IoT: An ACO-GA Approach," *2025 29th International Computer Conference, Computer Society of Iran (CSICC)*, Iran, 2025, pp. 1–5. doi: 10.1109/CSICC65765.2025.10967451.

SKILLS

Programming Languages:
Python, SQL, JavaScript, C++, C#, C

Data and ML:
Pandas, NumPy, Keras, TensorFlow, scikit-learn

Development and DevOps:
Git, Linux, Bash, Docker, CI/CD

Web:
React, HTML, CSS, Bootstrap, .NET

PROFESSIONAL EXPERIENCE

Tehran | Apr 2025 – Present
Data Scientist — Newtone (AI & FinTech Startup)

- Develop and maintain **data preprocessing, quality-validation and analytics pipelines** for large-scale banking datasets, **supporting Middle East Bank (Bankino)**.
- Conduct **credit-risk and customer-behavior analyses** to identify patterns in loan repayment, delinquency, and default, supporting data-driven improvements to scoring and risk assessment.
- Perform **cohort, segmentation and statistical analyses** to investigate relationships between customer behavior, financial characteristics, and **credit-risk outcomes**.

SELECTED PROJECTS

NoWaste — [GitHub]

- Developed a web application for reducing food waste, contributing to **frontend development, RESTful APIs and CI/CD pipelines**.

Anti-Spoofing Algorithm — [GitHub]

- Developed a **face liveness detection system**, implementing and comparing **SVC** and **VGG16-based deep-learning** approaches on the **CASIA-FASD dataset**.

Software Quality Measurement and Quantification — [GitHub]

- Developed a **quantitative software quality assessment** approach by translating qualitative quality criteria into measurable software metrics.

ACADEMIC EXPERIENCE

Honorary Member

- Voluntary work in Computer Engineering Scientific Association at Iran University of Science and Technology (CESA IUST) – Fall 2023

Teaching Assistant

- Systems Analysis and Design – Instructor: Dr. Mehdad Ashtiani – Fall 2024
- Data Transmission – Instructor: Dr. Marzieh Sheikhi – Spring 2024
- Artificial Intelligence – Instructor: Dr. Arash Abdi – Fall 2023
- Database Design – Instructor: Dr. Hossein Rahmani – Spring 2023
- Fundamentals of Computer Programming – Instructor: Dr. Marzieh Davoodabadi Farahani – Fall 2022

- Advanced Computer Programming – Instructor: Dr. Sauleh Eetemadi – Spring 2022
- Advanced Computer Programming – Instructor: Dr. Sauleh Eetemadi – Fall 2021

HONORS AND CERTIFICATES

Oct 2025	Credit Risk Modeling in Python, DataCamp
Oct 2025	ETL and ELT in Python, DataCamp
Oct 2025	Exploratory Data Analysis in Python, DataCamp
Oct 2024	Ranked within **the Top 10% among about 100** in my class in GPA at IUST
Jul 2022	Algorithms on Strings, Coursera
Mar 2022	Algorithms on Graphs, Coursera
Dec 2021	Data Structures, Coursera
Nov 2021	Algorithmic Toolbox, Coursera

LANGUAGES

Persian: Native Proficiency
English: Full Professional Proficiency IELTS (Academic): 7.0 (Overall score)